Do Now · retrieval

Compiler stages, linkers, loaders and libraries

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which program translates assembly mnemonics?

Q2

What does a compiler typically produce?

Q3

What is a runtime interpreter?

Q4

Simplify ¬(A∨B) with De Morgan's law, then evaluate A=0,B=1.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Lexical analysis identifies tokens, while syntax analysis checks grammar and generates a parse structure.

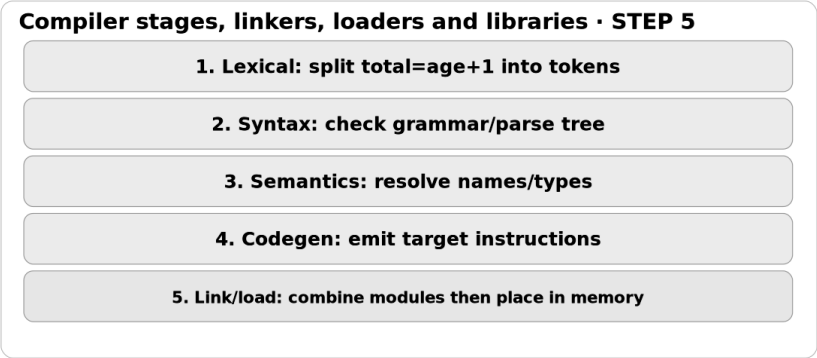
Core idea 2

Apply — Semantic analysis checks rules beyond grammar such as type compatibility or declared identifiers; optimisations can transform intermediate code.

Core idea 3

Evaluate — A linker combines object files and resolves symbols/libraries; a loader places the executable into memory for execution.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

lexer — Recognises source tokens.

parser — Checks grammatical structure.

semantic analysis — Checks program meaning and types.

linker — Resolves symbols across modules.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

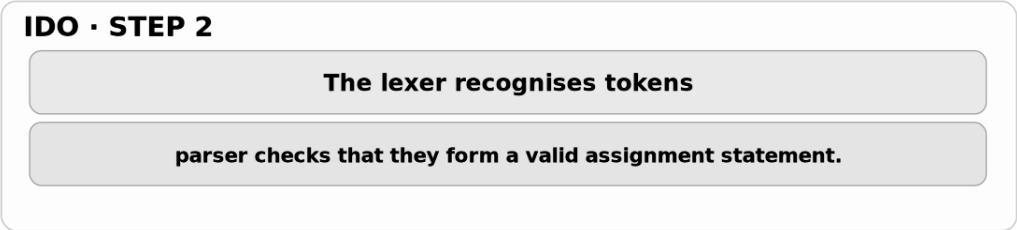
Problem

Analyse `total = count + 1;` into tokens and syntax structure.

Follow the worked method

1. Predict first — The lexer recognises tokens
2. Model answer — parser checks that they form a valid assignment statement.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
The linker finds its definition in the module/library and fixes up the reference before execution.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Distinguish lexical, syntax and semantic analysis. [6]

Question 2

Describe the roles of code generation, linking and loading. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A source file has valid syntax but calls undefined MethodX(). Explain which analyses or stages may detect failure, and why. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A parser guarantees that every variable in a syntactically valid program was declared.

Use key vocabulary from memory

1. Explain lexer in your own words.

2. Explain parser in your own words.

3. Explain semantic analysis in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.